

Practice Questions related to Final Exam - CSE422

Question 1: We have three Boolean variables: **P**, **Q**, and **R**. Below are the 8 observations:

P	Q	R
p	q	r
p	!q	r
!p	q	r
!p	!q	r
p	q	!r
p	!q	!r
!p	q	!r
!p	!q	!r

-Considering three variables P, Q, and R, construct a joint distribution table from above.

-Calculate $P(q \mid !r)$

Question 2: We construct a joint distribution table from three variables:

Device = {Laptop, Mobile}

User = {Student, Teacher}

Performance = {High, Low}

Device	User	Performance	Probability
Laptop	Student	High	0.12
Laptop	Student	Low	0.08
Laptop	Teacher	High	0.15
Laptop	Teacher	Low	0.05
Mobile	Student	High	0.2
Mobile	Student	Low	0.07
Mobile	Teacher	High	0.18
Mobile	Teacher	Low	0.15

Questions

1. Calculate $P(\text{Teacher})$.
2. What is $P(\text{High and Low})$?
3. Are **User** and **Performance** conditionally independent of each other, given **Device = Mobile**?

Question 3:

A company uses an email spam filter.

The filter correctly detects spam emails 90% of the time.

It incorrectly marks a legitimate email as spam 4% of the time.

In the company's inbox, 15% of all emails are spam.

If the filter marks an email as spam, what is the probability that it is actually not spam?

Question 4: A national park uses sensors and ranger observations to classify whether an animal sighting belongs to a rare species.

SightingID	TimeOfDay	Location	SoundDetected	RareSpecies
A01	Morning	Waterhole	Yes	Yes
A02	Night	Forest	No	No
A03	Evening	Forest	Yes	Yes
A04	Morning	Grassland	No	No
A05	Night	Waterhole	Yes	Yes
A06	Evening	Grassland	No	No
A07	Morning	Forest	Yes	Yes
A08	Night	Grassland	Yes	No

Using the Naive Bayes classifier, predict whether a new sighting belongs to a rare species given the following attributes:

TimeOfDay = Evening

Location = Waterhole

SoundDetected = Yes

Question 5: You are given the following dataset:

X	Y
1	5
3	12
5	15

Train a linear regression model with initial parameters $m=2$, $b=0$

1. Calculate the **Mean Squared Error (MSE)** for this model on the dataset.

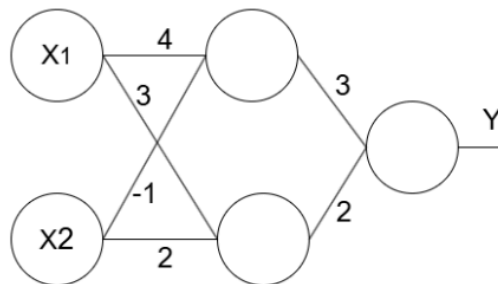
2. Perform **two iteration of gradient descent** to update both m and b, assuming the **learning rate = 0.05**.

Question 6:

X	Y
0.2	0
0.4	0
0.8	1

- If you fit the above data samples using $Y=mx+b$, where $m = 3$ and $b = 1$, calculate the amount of error using binary crossentropy loss function.
- Perform one iteration of gradient descent to update both m and b, assuming that the learning rate is 0.1

Question 7:



- If $X1 = 5$ and $X2 = 1$, apply forward propagation to calculate the value of Y. Consider that the hidden layer uses ReLU activation function while the output layer uses the sigmoid activation function.
- Mention the importance of activation functions in a neural network.

Question 8: Consider a **single neuron** (perceptron) with **one input** x and **one output** y

Initial weight: $w=0.4$

Initial bias: $b=0.1$

Learning rate: $\eta=0.1$

Perform one single iteration showing forward pass, weight update and back propagation